

Multiple Choice Questions: Logarithms

Mathematics Practice Module

December 30, 2025

1 Practice Questions

- Q1.** The value of $\log_{10} 0.001$ is:
A) 3 B) -3 C) 0.1 D) -2
- Q2.** If $\log_2 x = 5$, then the value of x is:
A) 10 B) 25 C) 32 D) 64
- Q3.** The value of $\log_5 1$ is:
A) 0 B) 1 C) 5 D) Undefined
- Q4.** $\log_a(mn)$ is equal to:
A) $(\log_a m)(\log_a n)$ B) $\log_a m + \log_a n$ C) $\log_a m - \log_a n$ D) $n \log_a m$
- Q5.** The value of $\log_3 81$ is:
A) 3 B) 9 C) 4 D) 27
- Q6.** If $\log_{10} 2 = 0.3010$, the value of $\log_{10} 80$ is:
A) 1.9030 B) 1.6020 C) 0.9030 D) 2.4080
- Q7.** The value of $\log_{\sqrt{2}} 32$ is:
A) 5 B) 10 C) 2.5 D) 20
- Q8.** If $\log_x(1/8) = -3/4$, then x is:
A) 16 B) 32 C) 64 D) 128
- Q9.** $\log_b a \cdot \log_c b \cdot \log_a c$ is equal to:
A) 0 B) abc C) 1 D) $\log(abc)$
- Q10.** The value of $\log_{10} 125 + \log_{10} 8$ is:
A) 2 B) 3 C) 1000 D) 100
- Q11.** If $\log_{10} x = y$, then $\log_{100} x^2$ is:
A) y^2 B) $2y$ C) y D) $y/2$
- Q12.** The value of $\log_2(\log_2(\log_2 16))$ is:
A) 0 B) 1 C) 2 D) 4
- Q13.** If $\log_y x = 3$ and $\log_z y = 2$, then the value of $\log_z x$ is:
A) 5 B) 1.5 C) 6 D) 9

- Q14.** $\log_{10} 5$ is equal to:
 A) $1 - \log_{10} 2$ B) $1 + \log_{10} 2$ C) $2 \log_{10} 2$ D) $0.5 \log_{10} 10$
- Q15.** The value of $7^{\log_7 11}$ is:
 A) 7 B) 1 C) 49 D) 11
- Q16.** If $\log_{10}(x+1) + \log_{10}(x-1) = \log_{10} 3$, then x is:
 A) 1 B) 2 C) ± 2 D) 4
- Q17.** The value of $\log_2 16 - \log_4 16$ is:
 A) 0 B) 1 C) 2 D) 4
- Q18.** Which of the following is not a valid base for a logarithm?
 A) 10 B) e C) 1 D) 0.5
- Q19.** If $\log_{10} 2 = a$ and $\log_{10} 3 = b$, then $\log_{10} 12$ is:
 A) $a + b$ B) $2a + b$ C) $a + 2b$ D) $2ab$
- Q20.** The value of $\log_8 128$ is:
 A) $7/3$ B) 16 C) $3/7$ D) 2
- Q21.** If $\log_4 x + \log_2 x = 6$, then x is:
 A) 4 B) 8 C) 16 D) 32
- Q22.** The value of $\log_{0.1} 10$ is:
 A) 1 B) -1 C) 0 D) 10
- Q23.** The logarithmic form of $a^b = c$ is:
 A) $\log_a b = c$ B) $\log_c a = b$ C) $\log_a c = b$ D) $\log_b c = a$
- Q24.** If $\log_{10} x = -2$, then x is:
 A) 0.01 B) 0.02 C) 0.001 D) 100
- Q25.** The value of $\log_3 27\sqrt{3}$ is:
 A) 3.5 B) 4 C) 3 D) 4.5
- Q26.** If $\log_2(x^2 - 1) = 3$, then x is:
 A) 3 B) ± 3 C) 9 D) $\pm\sqrt{7}$
- Q27.** The value of $(\log_3 5)(\log_{25} 27)$ is:
 A) 1.5 B) 1 C) 2 D) 0.75
- Q28.** If $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$, the number of digits in 3^{12} is:
 A) 5 B) 6 C) 7 D) 8
- Q29.** The value of $\log_5(0.008)$ is:
 A) -2 B) -3 C) 3 D) 2
- Q30.** $\log_b a$ is equal to:
 A) $\frac{\log a}{\log b}$ B) $\log(a - b)$ C) $\frac{\log b}{\log a}$ D) $\log a - \log b$

2 Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	B	13	C	19	B	25	A
2	C	8	A	14	A	20	A	26	B
3	A	9	C	15	D	21	C	27	A
4	B	10	B	16	B	22	B	28	B
5	C	11	C	17	C	23	C	29	B
6	A	12	B	18	C	24	A	30	A

3 Key Concept Hints

- **Basic Definition:** $\log_b x = y \iff b^y = x$.
- **Change of Base:** $\log_b a = \frac{\log_k a}{\log_k b}$. This is vital for Q9 and Q30.
- **Product/Quotient Laws:** $\log(mn) = \log m + \log n$ and $\log(m/n) = \log m - \log n$.
- **Power Law:** $\log(m^n) = n \log m$.
- **Common Bases:** Base 10 is common ($\log$), and base e is natural ($\ln$). A base must be positive and not equal to 1.
- **Calculating Digits (Q28):** The number of digits in x is $\lfloor \log_{10} x \rfloor + 1$.